

Review and critical assessment of dislocation loop analyses on EUROFER 97

C. Dethloff*, E. Gaganidze, J. Aktaa

Karlsruhe Institute of Technology (KIT), Institute for Applied Materials, Hermann-von-Helmholtz-Platz 1, Eggenstein-Leopoldshafen 76344, Germany



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ABSTRACT

The understanding of microstructural defects behavior after neutron irradiation is crucial for assessing the applicability of reduced activation ferritic/martensitic (RAFM) steel EUROFER 97 in future fusion reactors. Formation and evolution of dislocation loops is believed to play the major role in material's hardening under neutron irradiation. In this work, transmission electron microscopy (TEM) data on dislocation loop size distribution is provided after different irradiation campaigns to determine the role of neutron dose on the dislocation loop evolution. A comparison of investigations on dislocation loop behavior and appearance yield considerable differences. For a conclusive interpretation, this work reviews available data, and possible reasons for the observed differences are discussed. Recommendation for future TEM investigation are given.

Introduction

Neutron irradiation induced microstructural defects deteriorate mechanical properties of structural materials. Even though RAFM steels like the European variant EUROFER 97 are especially designed for withstanding the harsh environment in future fusion reactors, they still suffer from low temperature hardening and embrittlement which limit their application.

Dislocation loops, which form from interstitial atoms under neutron irradiation induced displacement cascades, are held responsible for the dominant contribution to hardening of EUROFER 97. Therefore, microstructural analysis of dislocation loops is the key for understanding irradiation effects and subsequent changes in mechanical properties. TEM is a powerful tool to analyze microstructural defects like dislocation loops since it provides a direct observation of the respective defect based on its diffraction contrast. Unfortunately, TEM has several uncertainties that have to be addressed when analyzing dislocation loop types, densities and size distributions. Analyses are affected by the lateral resolution of TEM, which is limited to defects larger than ~ 1 nm. While larger dislocation loops show a loop-like contrast, smallest loops only appear as black dots in TEM. Experimental results may vary depending on resolution and quality of the prepared samples. Further, at highest resolution the investigated sample volume is small i.e. statistics on defect densities are quite low and less reliable.

In this work new results on the dislocation loop microstructure in neutron irradiated EUROFER 97 are compared to previous

investigations after SPICE (15dpa, 300 °C, HFR), WTZ 01/577 (15dpa, 330 °C, BOR-60) and ARBOR1 (32dpa, 330 °C, BOR-60) irradiation experiments. Discrepancies in loop mean diameters and densities as well as their division in $a_0 \langle 100 \rangle$ and $a_0/2 \langle 111 \rangle$ type loops are observed in different studies under similar neutron irradiation conditions, and impede straightforward and conclusive findings. Possible reasons for these differences are considered including experimental and analytical conditions and sample preparation. A systematic analysis of existing microstructural data is performed and will contribute to deeper understanding of underlying irradiation effects.

Experimental

The basic material used in this work is the RAFM steel EUROFER 97 (rolled plate material, heat 83,697) produced by Böhler Austria GmbH with a composition of 8.91 Cr 1.08 W 0.48 Mn 0.20 V 0.14 Ta 0.006 Ti 0.12 C (wt.%, Fe balance) [1]. The material was delivered in a normalized (980 °C for 0.5 h) and tempered (760 °C for 1.5 h) condition. Several types of mechanical testing specimens were neutron irradiated in the irradiation experiments SPICE [2,3], WTZ 01/577 [4] and ARBOR1 [5]: the corresponding irradiation specifications are shown in Table 1.

TEM samples were manufactured in the Hot Cells at the Fusion Materials Laboratory (FML) of KIT from undeformed parts of irradiated EUROFER 97 impact test specimens with a cross-sectional area of 3×4 mm². By using a cutting wheel slices with thicknesses of about

Abbreviations: BF, Bright-Field; CBED, Convergent Beam Electron Diffraction; dpa, displacements per atom; EDS, Energy Dispersive X-ray spectroscopy; EELS, Electron Energy Loss Spectroscopy; EFTEM, Energy Filtered Transmission Electron Microscopy; FML, Fusion Materials Laboratory; HAADF, High Angle Annular Dark-Field; RAFM, Reduced Activation Ferritic / Martensitic; STEM, Scanning Transmission Electron Microscopy; TEM, Transmission Electron Microscopy; WBDF, Weak-Beam Dark-Field; ZA, Zone axis

* Corresponding author.

E-mail address: Christian.Dethloff@kit.edu (C. Dethloff).

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Nomenclature

a_0	lattice constant
g	diffraction vector
b	Burgers vector

150 μm to 200 μm were prepared and subjected to electrolytic polishing in a solution of 20% H_2SO_4 + 80% CH_3OH at room temperature with a Tenupol-5 jet polisher. Afterwards, in order to minimize radioactivity and also the influence of the magnetic sample on the electron beam, discs of 1 mm in diameter including the electron-transparent region were punched out. A foldable copper net carried the 1 mm sample and was used for examination in the TEM.

TEM investigations were performed at 200 kV on a high resolution FEI Tecnai G² F20 X-TWIN microscope equipped with a post-column GIF Tridiem energy filter and located in the Hot Cells at FML. Prior to each investigation the surface contamination of the TEM specimens was reduced by applying a plasma cleaning treatment of 10 min with air plasma. In TEM mode, all images were recorded with the GIF 2k camera and zero-loss filtered with an energy slit of 15 eV to improve contrast. In STEM, nanoprobe mode was used resulting in a probe diameter of approximately 1.5 nm, and images were processed by a HAADF detector. Elemental analysis by EDS was performed. For defect density evaluation the foil thickness of the investigated regions was determined by EELS log-ratio method [6] and CBED [7,8], and the recorded CBED patterns were analyzed using a DigitalMicrograph TM script [9].

Results

Dislocation loop analysis after WTZ irradiation (15 dpa in BOR-60) is performed to complete existing data on EUROFER 97. Only defects with clearly identified loop contrast, where the circularly shaped loops [10] appear elliptical due to projection or show a double arc contrast when oriented edge-on, are included and analysis is performed following the approach in [11]. The major axis of an elliptical or double arc loop contrast were taken as a measure of the dislocation loop size. A part of the investigated area is shown in Fig. 1; results are listed in Table 2 along with previous data for comparison. For a ZA of [01–2] and $g(3.1\text{ g})$ WBDF conditions, $g = \{121\}$ and $g = \{-200\}$ were analyzed, resulting in dislocation loop densities of $6.4 \times 10^{21}/\text{m}^3$ and $3.6 \times 10^{21}/\text{m}^3$, and mean sizes of 5.5 nm and 3.6 nm, respectively. Using the statistical method for Burgers vector determination [12], total loop density and mean size was determined to $6.9 \times 10^{21}/\text{m}^3$ and 6.1 nm, respectively, with a $\langle 100 \rangle$ loop fraction of 72%.

In addition, existing TEM micrographs from investigations by Weiß [13] after ARBOR1 irradiation (32 dpa, 330 °C) were re-evaluated taking into account only clearly recognizable dislocation loops. These results are also shown in Table 2, the total loop density and mean size was determined to $8.3 \times 10^{21}/\text{m}^3$ and 5.1 nm, respectively, with a $\langle 100 \rangle$ loop fraction of 27%. A direct comparison with data from [13], where the total dislocation loop densities were subsequently adapted using the statistical method for Burgers vector determination as previously published in [11], yield good agreement of data for 15 dpa and

32 dpa. The density of recognizable dislocation loops shows a slight increase with dose.

Discussion

Through assessing the literature data on dislocation loop analyses in EUROFER 97 after HFR and BOR-60 neutron irradiation [11,13,14], inconsistency of existing dislocation loop data is observed with respect to loop density, mean sizes and $\langle 100 \rangle / \langle 111 \rangle$ loop type ratio. Three major discrepancies attract attention: first, the defect density varies by one order of magnitude for the same or similar irradiation conditions; second, the mean defect size varies between 3 nm and 14 nm; third, the $\langle 100 \rangle$ loop frequency varies between 0% and 77%.

An error evaluation of the measured variables has been performed. For thickness determination by both EELS and CBED, an uncertainty of 10% [6] is assumed which directly influences the measured defect densities. For the error determination of the mean defect size two influencing factors are considered: the statistical error can be addressed by the standard error of the mean, while the TEM micrograph resolution (in nanometer per pixel) provides another source for the error. For a TEM magnification of 130,000 $\times$, as it is used in this work, an error of 0.2 nm is estimated for the differentiation of defects from the surroundings. The larger of these two errors is assumed as total error of the mean size, as proposed in [15]; Table 2 shows the estimated error magnitudes.

The main reason for density variation in literature is due to different defect definition criteria. As can be seen from results in Table 2 and is plotted in Fig. 2, clearly identifiable loops and black dots add up to the total defect density. Furthermore, Fig. 1a exemplarily displays the heterogeneous (“cloud-like”) loop distribution inside grains: while regions show a higher loop density, other parts of the grain show less visible loop contrast. Pre-existing microstructural defects, like the dislocation network from material fabrication, local variation of chemical composition (main effect of carbon) [16], and so forth might act as nucleation and trapping sites [17] for dislocation loops and could be responsible for the heterogeneous spatial distribution of dislocation loops after irradiation. In Fig. 3a, in addition and directly comparable to the base image of Fig. 1a, dislocation loops, which seem to arrange to straight lines, are marked with yellow lines, while visible line dislocations are marked with red lines. An estimation of the line (defect) density yield $8.2 \times 10^{13}/\text{m}^2$ for lined-up loops and $1.5 \times 10^{13}/\text{m}^2$ for visible line dislocations. In comparison, Fig. 3b shows the dislocation network in the unirradiated case, where the line dislocation density was determined to $1.5 \times 10^{14}/\text{m}^2$ by method of intersecting grid. Under the assumption that the lined-up dislocation loops follow line dislocations, the values of the line dislocation density are comparable in the unirradiated case and after irradiation to 15 dpa at 330 °C.

The heterogeneous spatial distribution of dislocation loops, in general, leads to an overestimation of the density because the observer preferentially analyses regions where large amounts of loops are visible. Thus, the investigated area has to be increased to minimize that effect. The image resolution, that is the combination of microscope magnification, camera resolution, sample quality and thickness, is crucial for distinct detection of dislocation loops. Only well prepared samples and a well-adjusted microscope allow for clear identification of loops by minimization of superimposed diffraction contrast, and accurate separation of loops from black dots and other defects. The preceding reasons also lead to an overestimation of the detected mean loop size. Underestimation of the loop density due to partial invisibility is resolved by using the statistical method for Burgers vector analysis [12].

In [14], only $\langle 111 \rangle$ type loops were detected with a substantially higher mean diameter of 14 nm. To find a reason for this observation, the following consideration should be taken into account. In general, as was shown in [11], $\langle 111 \rangle$ loops are larger in size. Under the assumption of a lower microscope magnification in [14] than e.g. in [11] and the focusing of a relatively low sample area close to the larger $\langle 111 \rangle$ loops,

Table 1
Details on irradiation experiments relevant for this work.

Irradiation experiment	ARBOR1	WTZ 01/577	SPICE
Reactor	BOR-60	BOR-60	HFR
	Dimitrovgrad	Dimitrovgrad	Petten
Neutron flux _{E>0.1 MeV} ($10^{18} \text{ m}^{-2}\text{s}^{-1}$)	18	18	4.0
Neutron flux _{thermal} ($10^{18} \text{ m}^{-2}\text{s}^{-1}$)	–	–	1.4
Dose (dpa)	32	15	15
Irradiation temperature (°C)	330	330	300

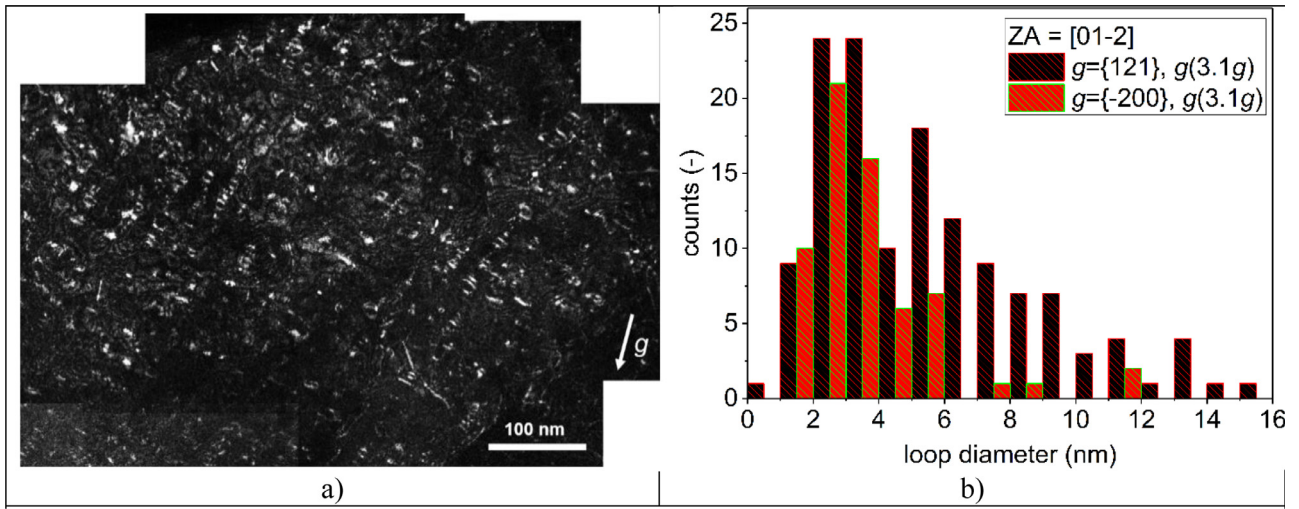


Fig. 1. WBD analysis of dislocation loops after WTZ irradiation (15dpa, 330 °C). A part of the investigated volume is shown in (a) for ZA = [01-2], $g = \{121\}$. In (b), the dislocation loop size distribution is presented.

Table 2

Summary of results from TEM analyses on dislocation loops in RAFM steel EUROFER 97 and F82H after neutron irradiation. Error in sizes is given where available and is the larger of the standard error of the mean and pixel resolution of the image; error in number densities (not shown) is estimated to be about 10% due to uncertainties in the TEM foil thickness measurement.

Material	Neutron irradiation	Dose (dpa)	T_{irr} (°C)	Defect Types	Density ($10^{21}/m^3$)	Mean size (nm)	Standard error of mean (nm)	Error of size due to image pixel resolution (nm)	$\langle 100 \rangle$ number fraction (%)	References
EUROFER 97	HFR	15	300	loops	6.1	3.3 ± 0.2	0.2	0.2	77	[11]
EUROFER 97	HFR	15	300	loops	4.0	14.0	–	–	0	[14]
EUROFER 97	BOR-60	15	330	loops	6.9	6.1 ± 0.3	0.3	0.2	72	this work
EUROFER 97	BOR-60	15	330	loops + black dots	20.0	3.4 ± 0.2	0.2	0.2	45	[11], re-calculated from [13]
EUROFER 97	BOR-60	32	330	loops	8.3	5.1 ± 0.3	0.2	0.3	27	this work, re-evaluation of micrographs from [13]
EUROFER 97	BOR-60	32	330	loops + black dots	24.0	4.8 ± 0.3	0.1	0.3	31	[11], re-calculated from [13]
F82H	HFR	9.2	310	–	28.0	6.9	–	–	–	[18]
F82H	HFR	2.9	331	–	54.0	16.3	–	–	–	[18]

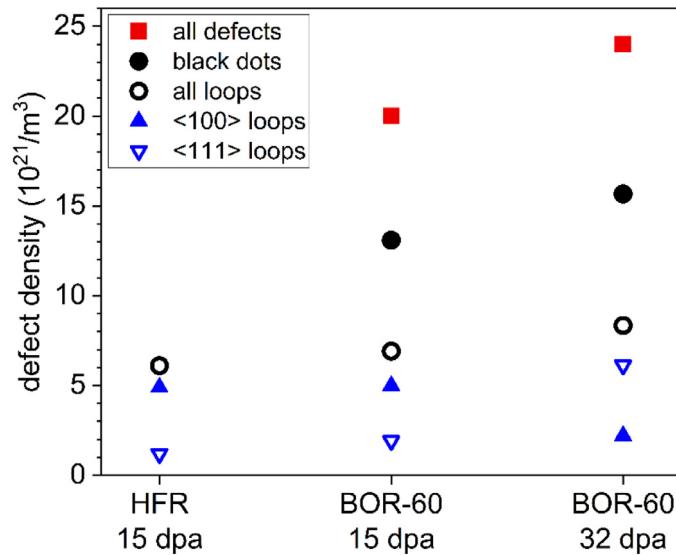


Fig. 2. Summary of TEM results on dislocation loop analysis from this work and [11,13]. New results allow differentiation of different defect types into loops and black dots, and the observed densities of $\langle 100 \rangle$ and $\langle 111 \rangle$ type loops are presented.

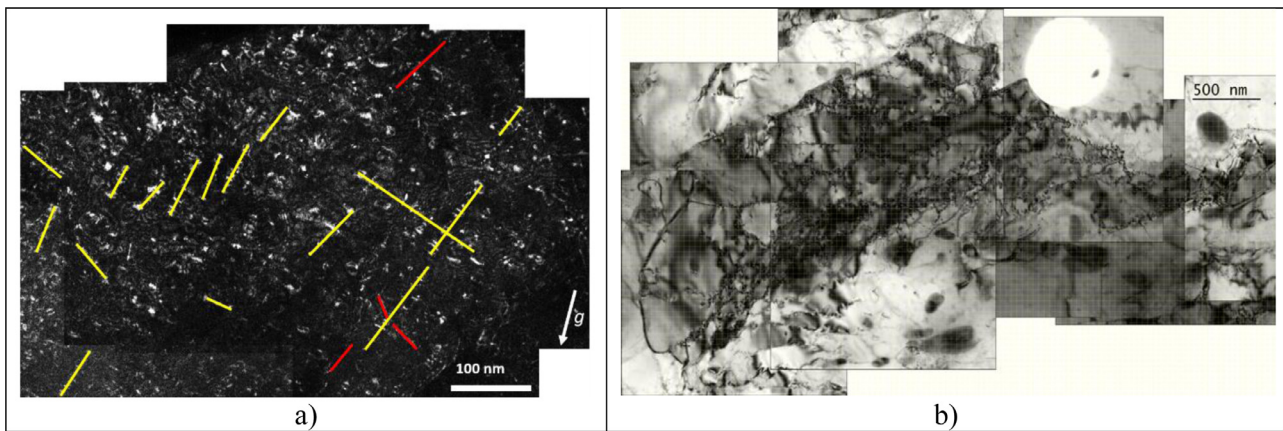


Fig. 3. (a) Based on Fig. 1a under WBDF condition $g(3.1\text{ g})$, $g = \{121\}$, the heterogeneous lateral distribution of dislocation loops is observed after WTZ irradiation. Lined-up dislocation loops are marked by yellow lines, visible line dislocations by red lines. (b) Determination of the dislocation density in unirradiated EUROFER 97 by intersecting grid (gray overlay) method in TEM BF and tilted to a ZA of $[012]$.

smaller $\langle 100 \rangle$ type loops might be missed and may be excluded from the analysis. As mentioned before, these reasons may lead to an over-estimation of dislocation loop sizes, and the quantity of smaller loops, which are preferably of $\langle 100 \rangle$ type, is highly underestimated.

Even when putting aside results from [14], where exclusively $\langle 111 \rangle$ loops were observed with a considerably larger mean diameter of 14 nm, and focusing on results from this work and [13,11], the frequency of $\langle 100 \rangle$ and $\langle 111 \rangle$ type loops reverses between 15dpa and 32dpa in BOR-60. Fig. 2 summarizes dislocation loop analysis. For the $\langle 100 \rangle / \langle 111 \rangle$ loop ratio, the inconsistency remains unclear, and further investigation is necessary.

Data on F82H after 9.2dpa at 310 °C in HFR [18] showed comparable results regarding a total defect density and a mean defect size. After 2.9dpa at 331 °C, however, a substantially larger mean defect size of 16.3 nm was observed, which was supposed to originate from the higher irradiation temperature. In [14] it was shown, that for 350 °C EUROFER 97 showed a larger mean loop size, which supports the assumption of a change in defect evolution of dislocation loops between approximately 330 °C and 350 °C, even though for 330 °C in BOR-60 this effect could not be observed.

Conclusion

New results on dislocation loop appearance in neutron irradiated EUROFER 97 after 15dpa at 330 °C in BOR-60 reactor were presented. Obtained data provides good agreement with our preceding results and partly solves inconclusive findings concerning dislocation loop and black dot density and mean size. It is shown that the particular defect density strongly depends on the defect definition criteria, and thus the possibility of separation of clearly visible dislocation loops from black dots. Uncertainties of TEM investigation were discussed which leads to over- and underestimation of defect density and mean sizes; in particular the following consideration should be treated with great care:

- Since we want to identify defects with sizes close to the resolution limit of TEM, the microscope magnification has to be sufficiently high, and the microscope adjustment and sample preparation has to be optimized.
- A sufficient sample volume has to be analyzed to avoid over-estimation of defect sizes and densities.
- The use of the statistical method for Burgers vector analysis avoids underestimation of defect densities.
- Accurate and complete data on the investigation approach must be

provided, that includes e.g. defect definition criteria, total numbers of counted defects, investigated sample volume, etc.

A partial line-up of dislocation loops along possible line dislocations is observed after irradiation, and the estimated dislocation density is comparable to the unirradiated case of about $10^{14}/\text{m}^2$. For the observed and varying $\langle 100 \rangle / \langle 111 \rangle$ loop ratios, further investigation is necessary since the loop appearance is not satisfactorily understood.

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References

- [1] E. Gaganidze, H.-C. Schneider, B. Dafferner, J. Aktaa, *J. Nucl. Mater.* 355 (2006) 83–88.
- [2] E. Gaganidze, B. Dafferner, H. Ries, R. Rolli, H.-C. Schneider, J. Aktaa, *Wissenschaftliche Berichte, FZKA 7371*, Forschungszentrum Karlsruhe (2008).
- [3] E. Materna-Morris, A. Möslang, R. Rolli, H.-C. Schneider, *J. Nucl. Mater.* 386–388 (2009) 422–425.
- [4] C. Petersen, J. Aktaa, E. Diegele, E. Gaganidze, R. Lässer, E. Lucon, E. Materna-Morris, A. Möslang, A. Povstnyanko, V. Prokhorov, J. Rensman, B. van der Schaaf, H.-C. Schneider, *Mechanical properties of reduced activation ferritic/martensitic steels after european reactor irradiations*, 21st IAEA Fusion Energy Conf., Chengdu, China, October 16–21, FT/1-4Ra, 2006.
- [5] C. Petersen, V. Shamardin, A. Fedoseev, G. Shimansky, V. Efimov, J.W. Rensman, *J. Nucl. Mater.* 307–311 (2002) 1655–1659.
- [6] H.-R. Zhang, R.F. Egerton, M. Malac, *Micron* 43 (2012) 8–15.
- [7] P.M. Kelly, A. Jostsons, R.G. Blake, J.G. Napier, *Phys. Status Solidi (a)* 31 (1975) 771–780.
- [8] S.M. Allen, *Philos. Mag. A* 43 (1981) 325–335.
- [9] V. Hou, *Microsc. Microanal.* 10 (2004) 1380–1381.
- [10] B. Yao, D.J. Edwards, R.J. Kurtz, *J. Nucl. Mater.* 434 (2013) 402–410.
- [11] C. Dethloff, E. Gaganidze, J. Aktaa, *Nuclear Mater Energy* 9 (2016) 471–475.
- [12] A. Prokhorov, B. Décamps, A. Ramar, R. Schäublin, *Acta Mater.* 61 (2013) 6958–6971.
- [13] O.J. Weiß, E. Gaganidze, J. Aktaa, *J. Nucl. Mater.* 426 (2012) 52–58.
- [14] M. Klimenkov, E. Materna-Morris, A. Möslang, *J. Nucl. Mater.* 417 (2011) 124–126.
- [15] Z. Jiao, S. Taller, K. Field, G. Yeli, M.P. Moody, G.S. Was, *Microstructure evolution of T91 irradiated in the BOR60 fast reactor*, *J. Nucl. Mater.* (2018), <http://dx.doi.org/10.1016/j.jnucmat.2018.03.024>.
- [16] L. Morsdorf, C.C. Tasan, D. Ponge, D. Raabe, *Acta Mater.* 95 (2015) 366–377.
- [17] D. Terentyev, N. Anento, A. Serra, *J. Nucl. Mater.* 420 (2012) 9–15.
- [18] R. Schäublin, M. Victoria, *MRS Symp. Proc.* 650 (2001) R181–R186.